



Generative AI Applications in SE Lab 01

Submitted to

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Submitted by

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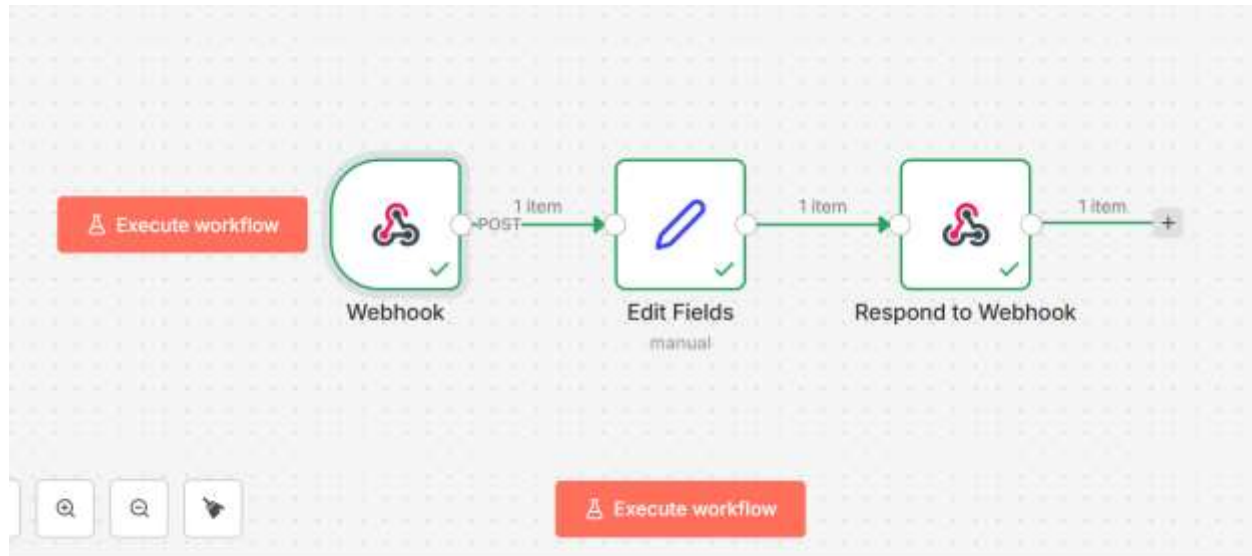
BSSE-VI-B

Objective

To install and configure n8n locally and build two practical workflows — a Webhook-based mini API and a scheduled office-hours logger.

ACTIVITY 1 — Workflow A: Webhook Echo API

Workflow Screenshot:



What I Did: Built a workflow that receives a POST request via Webhook, extracts name, message, and timestamp using a Set node, and returns a clean JSON response.

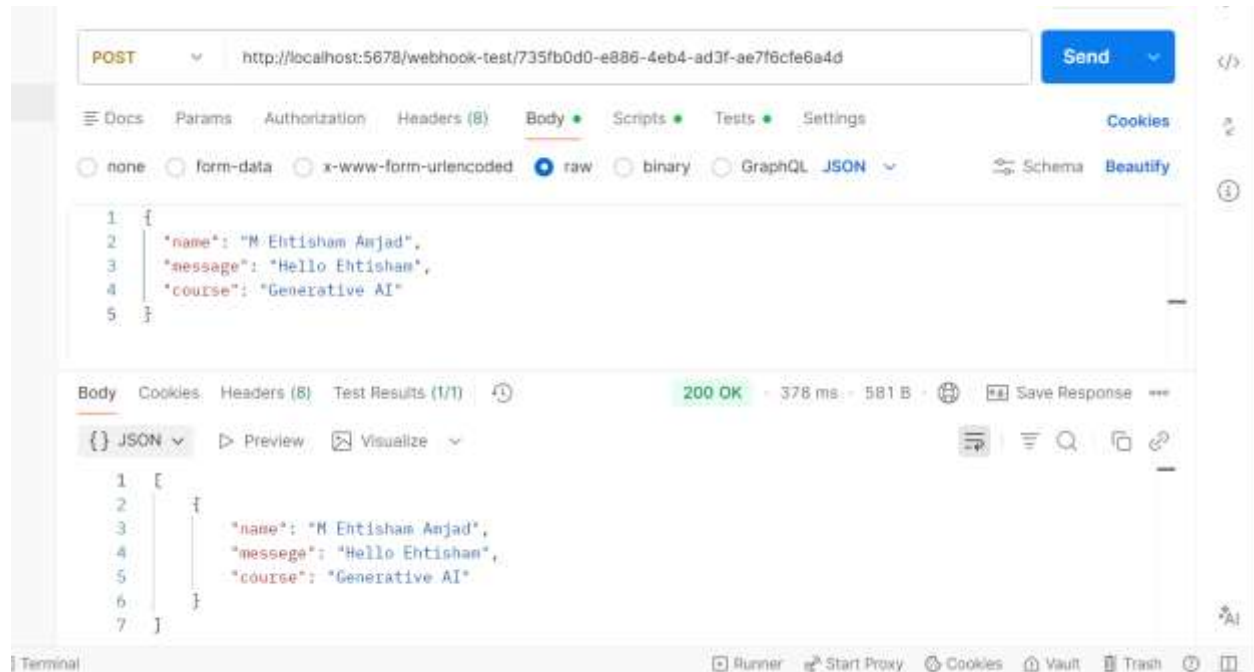
Node Settings:

Node	Settings
Webhook	Method: POST, Path: lab01/echo, Respond: Using Respond to Webhook Node
Edit Fields (Set)	name = \$json.body.name, message = \$json.body.message, receivedAt = \$now
Respond to Webhook	Code: 200, Body: First Incoming Item

Expected Output:

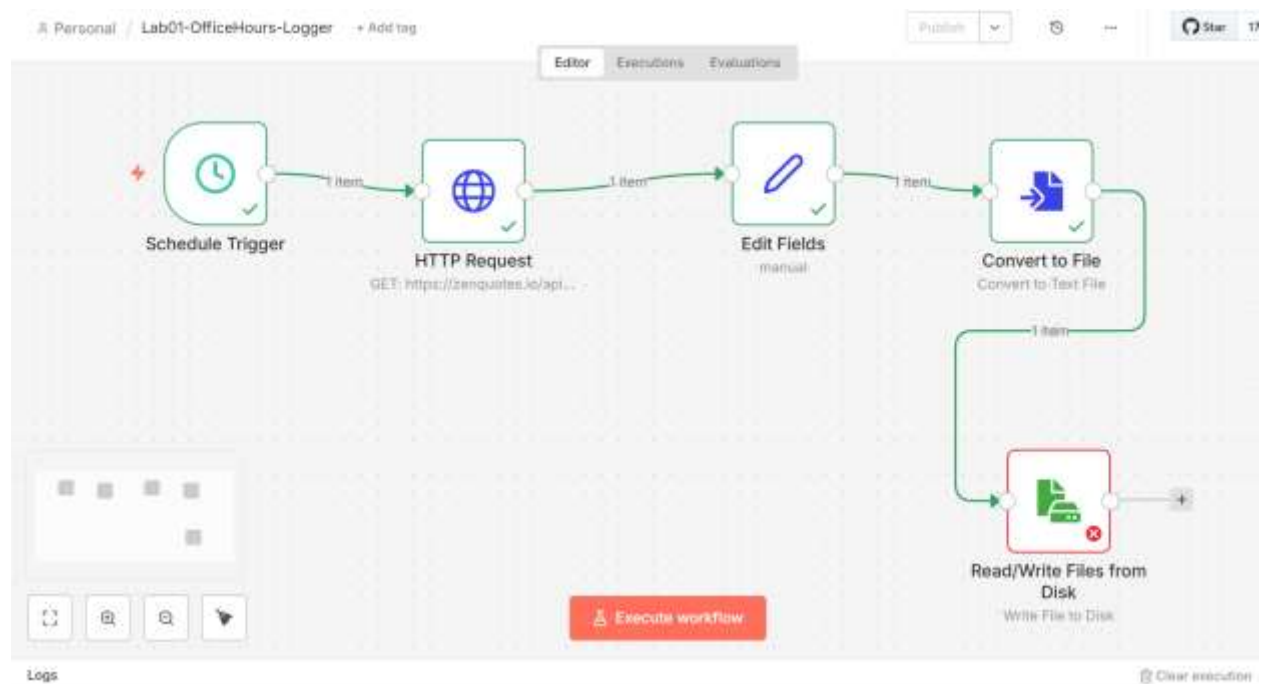
```
{"name":"M Ehtisham Amjad","message":"Hello Ehtisham","course":"Generative AI"}
```

Test Screenshot:



ACTIVITY 2 — Workflow B: Office-Hours Status Logger

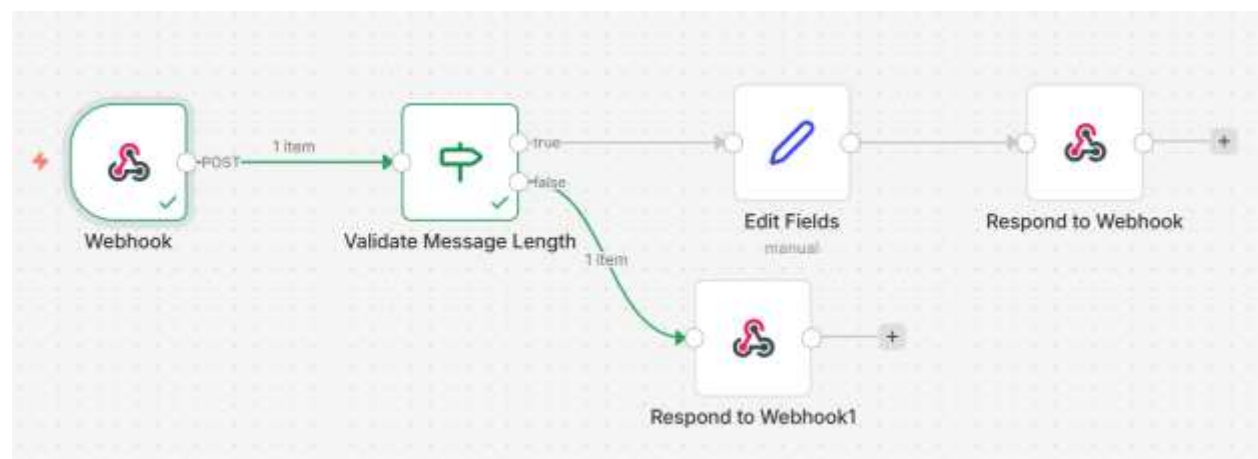
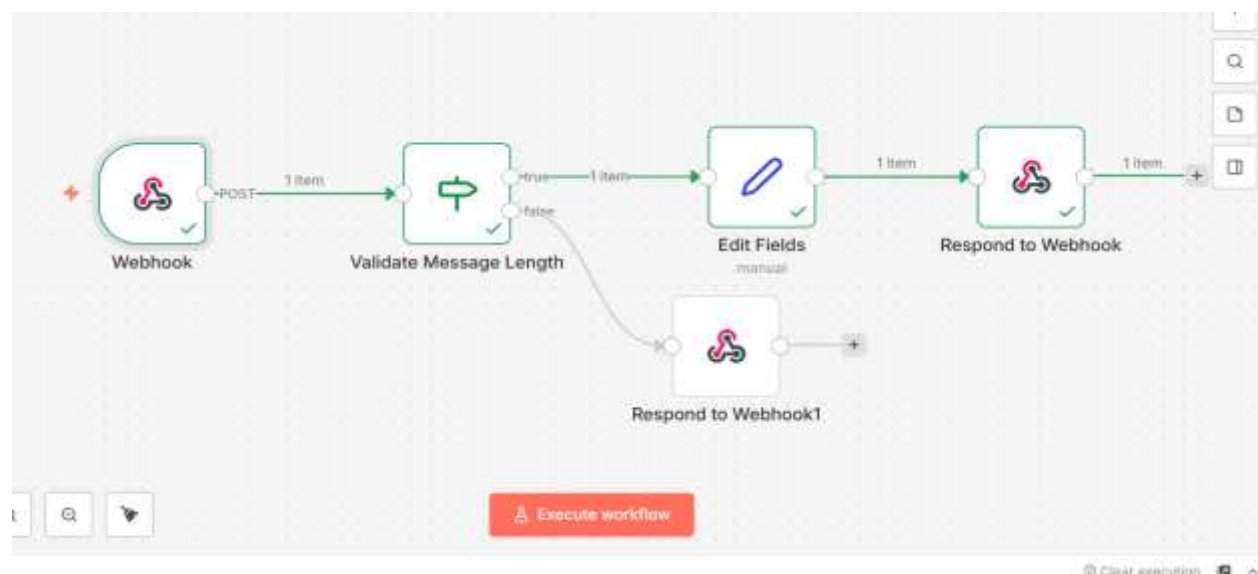
Workflow Screenshot:



What I Did: Built a scheduled workflow that runs Mon–Fri at 09:00, fetches a random quote from an API, and writes it to a local text file.

Node Settings:

Node	Settings
Schedule Trigger	Mode: Weekly, Days: Mon–Fri, Time: 09:00
HTTP Request	Method: GET, URL: https://zenquotes.io/api/random
Edit Fields (Set)	quote = \$json[0].q, author = \$json[0].a, loggedAt = \$now
Read/Write Files	Path: /tmp/quotes.txt, Append: ON

GRADED TASK 1 — Webhook-Based Input Validation (Marks: 5 | Difficulty: Low)**Workflow Screenshot:**

What I Did: Extended Workflow A by adding an IF node that checks if the message field is at least 5 characters. Valid messages return HTTP 200 accepted. Short messages return HTTP 400 rejected.

Node Settings:

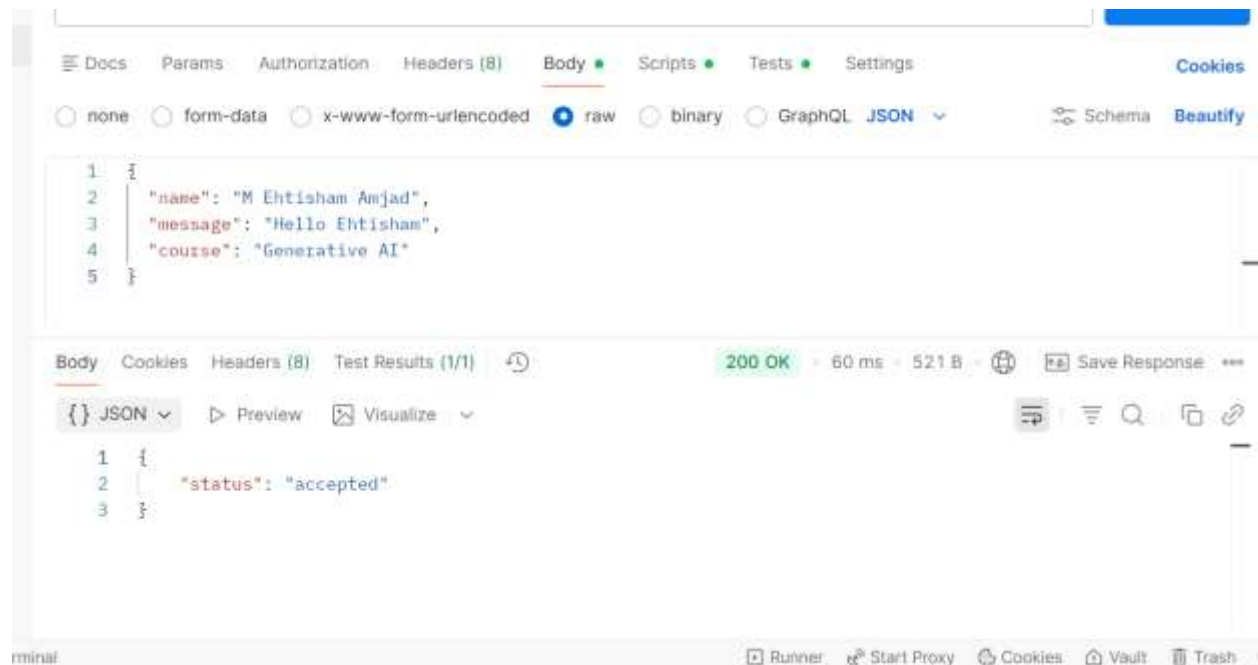
Node	Settings
Webhook	Method: POST, Respond: Using Respond to Webhook Node
IF Node	Condition: message length >= 5
Edit Fields	status = "accepted"
Respond to Webhook	Code: 200, Body: {"status": "accepted"}
Respond to Webhook1	Code: 400, Body: {"status": "rejected"}

Test Results:

Test	Input	Response Code	Response Body	Result
Valid message	{"message": "Hello World"}	200	{"status": "accepted"}	PASS
Invalid message	{"message": "Hi"}	400	{"status": "rejected"}	PASS

Test Screenshots:

Status (Accepted)



Status (Rejected)

```

2 | "name": "M Ehtisham Amjad",
3 | "message": "H",
4 | "course": "Generative AI"
5 | }

```

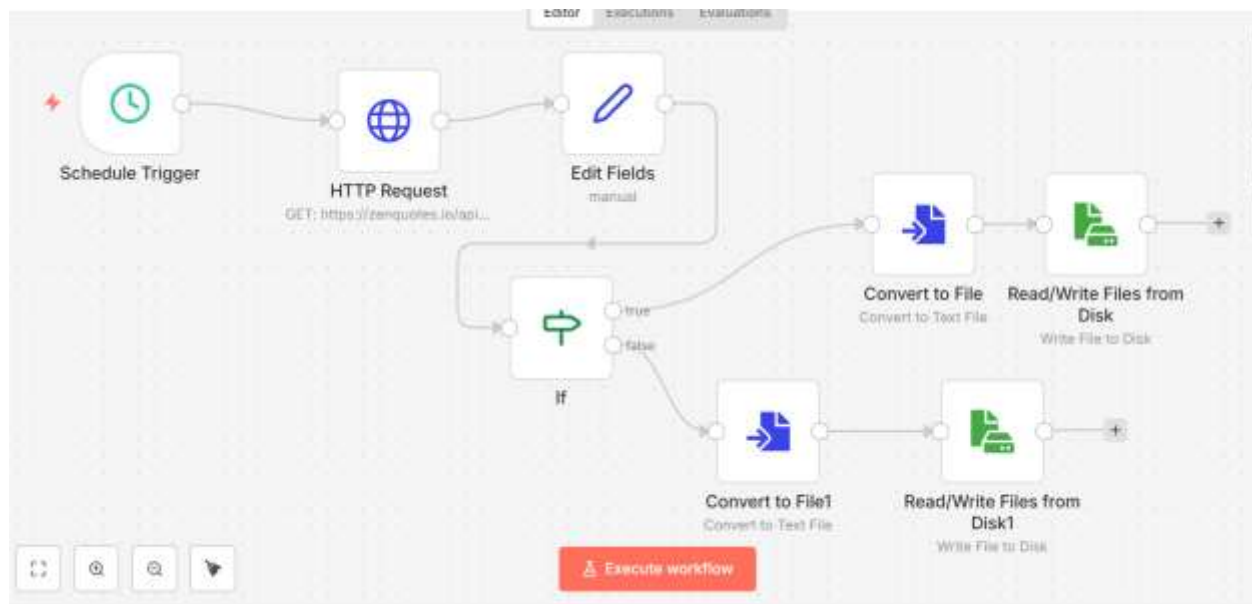
Body Cookies Headers (8) Test Results (1/1) ↻

{ } JSON ▾ ▶ Preview 🖼 Visualize ▾

```

1 | {
2 |   "status": "rejected"
3 | }

```

GRADED TASK 2 — Office-Hours Filter (IF Node) (Marks: 7 | Difficulty: Medium)**Workflow Screenshot:**

What I Did: Added an IF node to Workflow B that checks if the current hour is between 09:00 and 17:00. Within office hours the quote is logged to quotes.txt. Outside office hours "Skipped (outside office hours)" is written to skipped.txt.

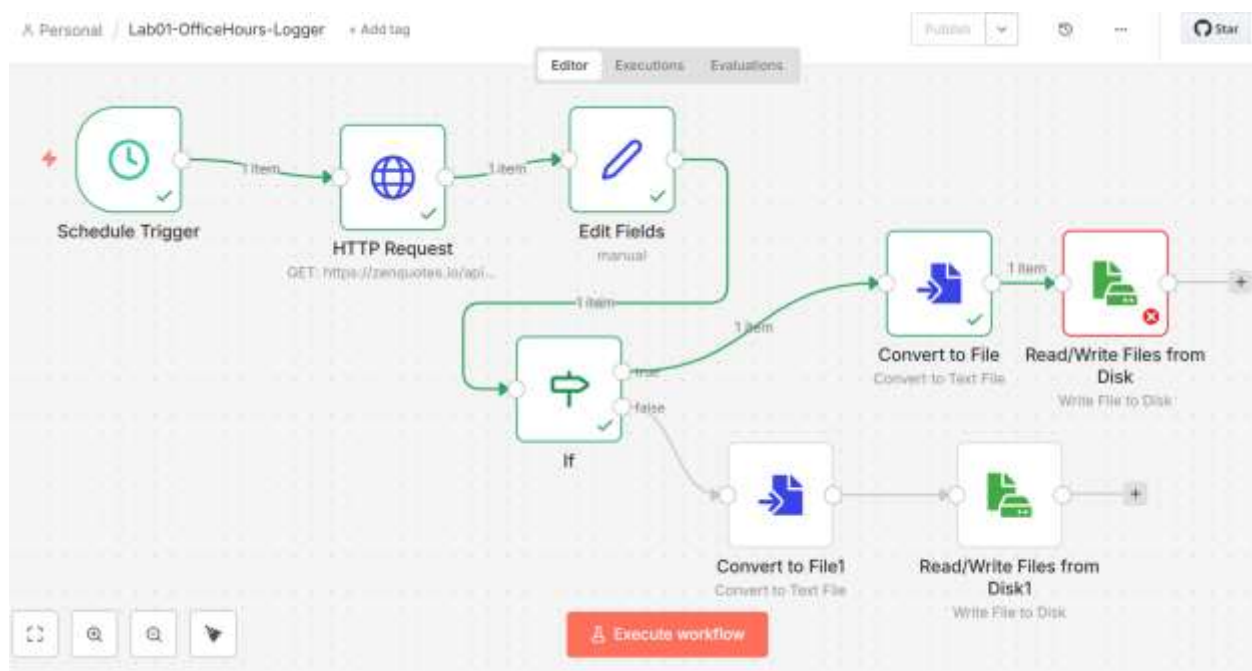
Node Settings:

Node	Settings
IF Node (Office Hours)	Condition 1: hour >= 9, Condition 2: hour < 17, Combined: AND
Convert to File	Text Input: quote, Output Field: data
Write quotes.txt	Path: /tmp/quotes.txt, Append: ON
Convert to File 2	Text: "Skipped (outside office hours)", Output Field: data
Write skipped.txt	Path: /tmp/skipped.txt, Append: ON

Test Results:

Test	Condition	Path Taken	Output File	Result
Within office hours	Current time 09:00–17:00	TRUE	quotes.txt updated	PASS
Outside office hours	Hour forced to 25 in IF node	FALSE	skipped.txt updated	PASS

Test Screenshots:



GRADED TASK 3 — Workflow Export and Documentation (Marks: 8 | Difficulty: Medium)

What I Did: Exported both workflows as JSON files using the three dots menu → Download option in n8n. Compiled all screenshots and node settings into this document.

Exported Files:

File	Workflow
Workflow_A_Task1.json	Webhook-Based Input Validation
Workflow_B_Task2.json	Office-Hours Logger with IF Node

Reflection

Working on this lab gave me a solid understanding of how n8n handles both event-driven and scheduled automation. Building Workflow A helped me understand how webhooks work as API endpoints and how conditional logic can be used to validate incoming data. Workflow B showed how scheduled triggers combined with IF nodes can create time-aware automations. The most challenging part was Docker networking — external API calls were blocked by default DNS, fixed by adding `--dns 8.8.8.8`. File writing inside Docker also required running the container as root user. Overall the lab gave me practical experience with real automation patterns I can apply in actual projects.

Summary

Task	Description	Marks	Status
Activity 1	Webhook Echo API	-	Completed
Activity 2	Office-Hours Logger	-	Completed
Task 1	Webhook Input Validation	5	Completed
Task 2	Office-Hours IF Filter	7	Completed
Task 3	Export and Documentation	8	Completed
Total		20	All Done

End Lab 01